

SMART FARMING OS



KrishiOS

Empowering rural agriculture through
AI, IoT & accessible technology

ফসল

AI

বাজার

মোবাইল

Team **Krishi OS**



Bangladesh · 2025



AI Advisory



IVR Voice



Solar IoT



11M Farmers

WebApp Link: <https://krishios.netlify.app/>

The Problem

Rural Farming in Bangladesh



Climate Uncertainty

Unpredictable weather patterns increase crop losses and risk for smallholder farmers.



Limited Internet Access

Connectivity gaps prevent smartphone-first solutions from reaching most farmers.



Literacy Barriers

Illiteracy and tech-unfamiliarity block adoption of modern digital methods.



Low Crop Productivity

Traditional practices and poor inputs reduce yields and household incomes.



No Real-Time Guidance

Farmers lack actionable, timely advice for planting and harvesting windows.

These problems combine to suppress incomes, increase vulnerability, and limit sustainable productivity.

Our Solution

The KrishiOS Ecosystem

One integrated, offline-capable system that brings AI insights and voice-first access to rural farms.



AI Farming Assistant

Localized recommendations for planting, nutrition, and pest management tailored to each field.



IVR Voice Support (Bangla)

Actionable audio alerts and two-way voice queries, no smartphone required.



Solar-Powered & Offline

Designed to operate with intermittent connectivity and limited power supply.



Smart Monitoring System

Low-power sensors continuously collect soil, moisture, and microclimate data.

WebApp Link: <https://krishios.netlify.app/>

How It Works

From field sensor to farmer action — in 4 simple steps

01



Field Sensors

Low-power IoT sensors collect soil moisture, nutrients & microclimate data from the farm.

02



Local Gateway

Edge gateway aggregates sensor data and syncs with the cloud when connectivity is available.

03



AI Analysis

Machine learning models process field data and generate crop-specific, localized recommendations.

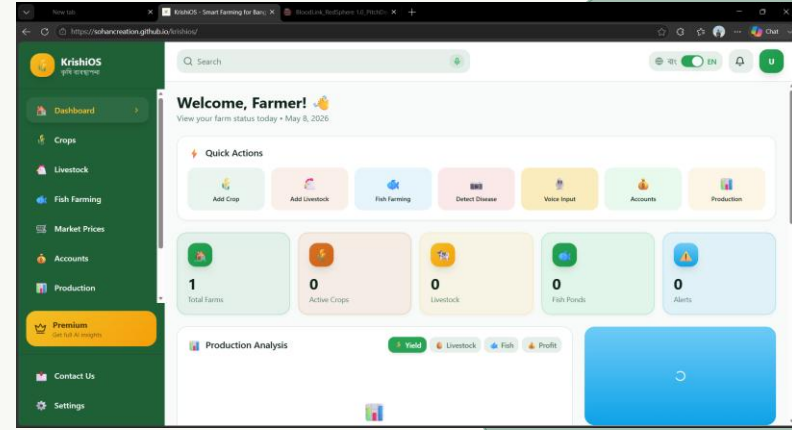
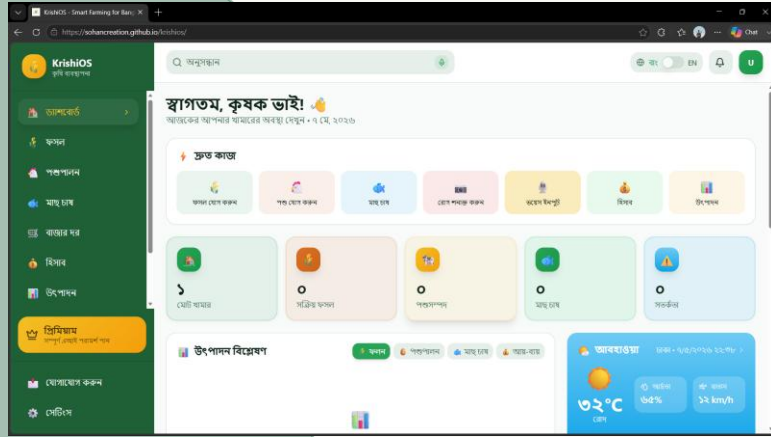
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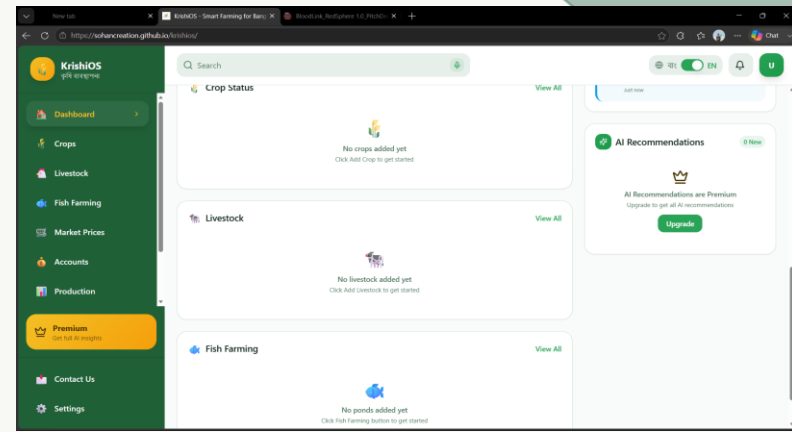
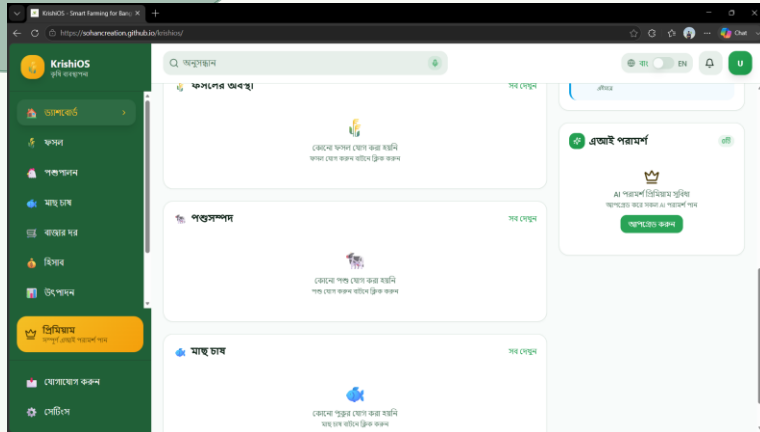
Farmer Delivery

Advice delivered via IVR voice call in Bangla, SMS alerts, or partner dashboard — no smartphone needed.

How It Works — Dashboard



KrishiOS Webapp Dashboard (Bangla & English)



কৃষিOS-এ স্বাগতম

আপনার কৃষি সহায়ক তৈরি করুন ~১ মিনিটে

অ্যাকাউন্ট তৈরি

প্রোফাইল ছবি যোগ করুন (ঐচ্ছিক)

পুরো নাম *

আপনার নাম

ইমেইল *

email@example.com

মোবাইল নম্বর *

01XXXXXXX

পাসওয়ার্ড *

.....

পুনরায় পাসওয়ার্ড *

.....

জেলা *

জেলা নির্বাচন করুন

উপজেলা *

উপজেলা নির্বাচন করুন

অ্যাকাউন্ট তৈরি করুন

ইতোমধ্যে অ্যাকাউন্ট আছে? লগইন করুন

কৃষিOS-এ স্বাগতম

আপনার কৃষি সহায়ক তৈরি করুন ~১ মিনিটে

কৃষকের ধরন

আপনি কী ধরনের কৃষক? (একাধিক নির্বাচন করা যাবে)

ফসল

সবদিপশু

পোল্ট্রি

মিশ্র

পরবর্তী →

Create an Account
Step by step based on your requirements

কৃষিOS-এ স্বাগতম

আপনার কৃষি সহায়ক তৈরি করুন ~১ মিনিটে

চাষের ধরন

আপনি কোন পদ্ধতিতে চাষ করেন?

জৈব

কেমিক্যাল

মিশ্র

← পিছনে

পরবর্তী →

কৃষিOS-এ স্বাগতম

আপনার কৃষি সহায়ক তৈরি করুন ~১ মিনিটে

সমস্যা চিহ্নিত

আপনার সবচেয়ে বড় সমস্যা কী? (একাধিক নির্বাচন করা যাবে)

কম ফলন

রোগ

বাজার মূল্য

বেশি খরচ

পানি সমস্যা

পশুর অসুখ

← পিছনে

সম্পন্ন করুন

কৃষিOS-এ স্বাগতম

আপনার কৃষি সহায়ক তৈরি করুন ~১ মিনিটে

খামারের তথ্য

আপনার মোট জমি/খামারের আকার কত? *

১ বিঘার কম

১-৫ বিঘা

৫-২০ বিঘা

২০+ বিঘা

শুধু খামার (জমি নেই)

জমি/খামার আপনার নিজের নাকি শিজ? *

নিজের

শিজ

মিশ্র

সেচ বা পানির উৎস কী? *

টিউবওয়েল

পুকুর/খাল

বৃষ্টি নির্ভর

জানি না

← পিছনে

পরবর্তী →

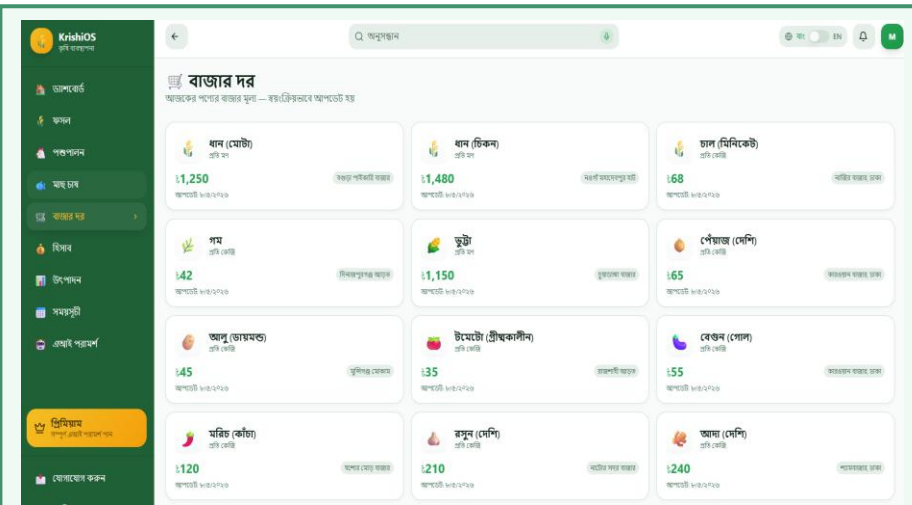
How It Works — Dashboard

Scheduling & Market Prices



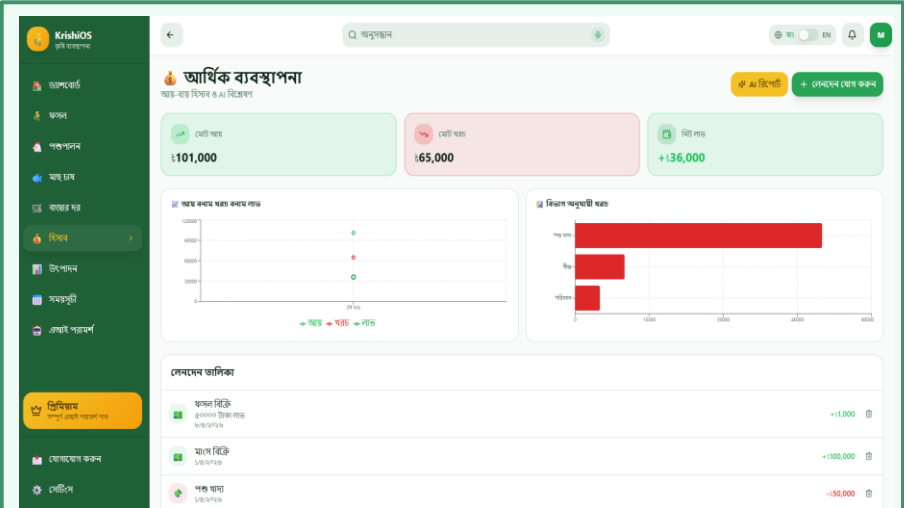
Smart Scheduling (📅 সময়সূচী ও কাজের তালিকা)

AI-assisted task planner that organizes farming activities (soil prep, seed sowing, pest control). Provides reminders, urgency markers, and personalized guidance to keep farmers on track.



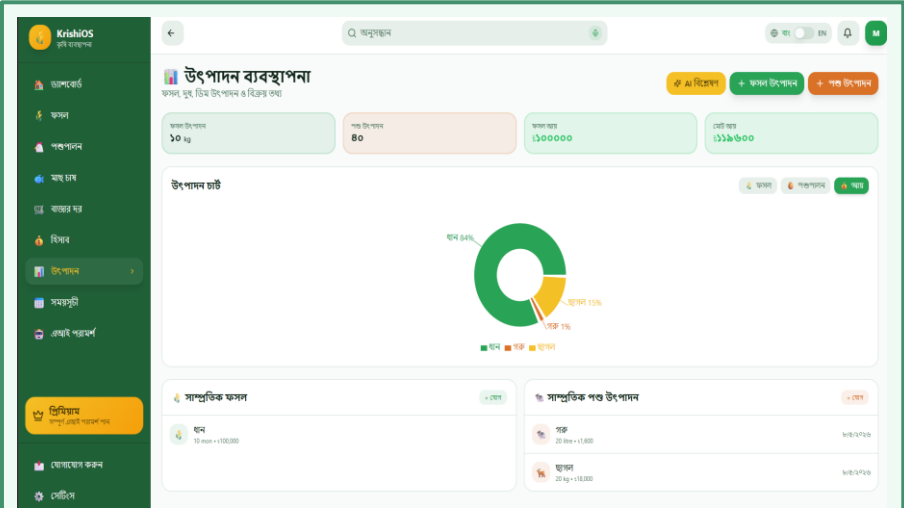
Market Price Updates (বাজার দর)

Displays real-time commodity prices (rice, wheat, vegetables, spices). Helps farmers make informed selling decisions, maximize profit, and adapt to market fluctuations.



Financial Management (আর্থিক ব্যবস্থাপনা)

Tracks farmer’s income, expenses, and net profit with AI-based financial analysis. Farmers can add transactions, generate AI reports, and monitor profitability in real time.



Production Management (উৎপাদন ব্যবস্থাপনা)

Records crop and livestock production data. Farmers can log yields, calculate income, and access AI analysis for productivity trends. Supports decision-making on crop cycles and livestock output.

How It Works — Step 1

AI Advisory Engine

01 AI-Powered Crop Advisory

The AI engine analyzes real-time sensor data, historical crop patterns, and regional weather forecasts to provide hyper-local farming advice.



Planting window recommendations based on soil & weather data



Pest & disease alerts with image-based detection (photo scan)



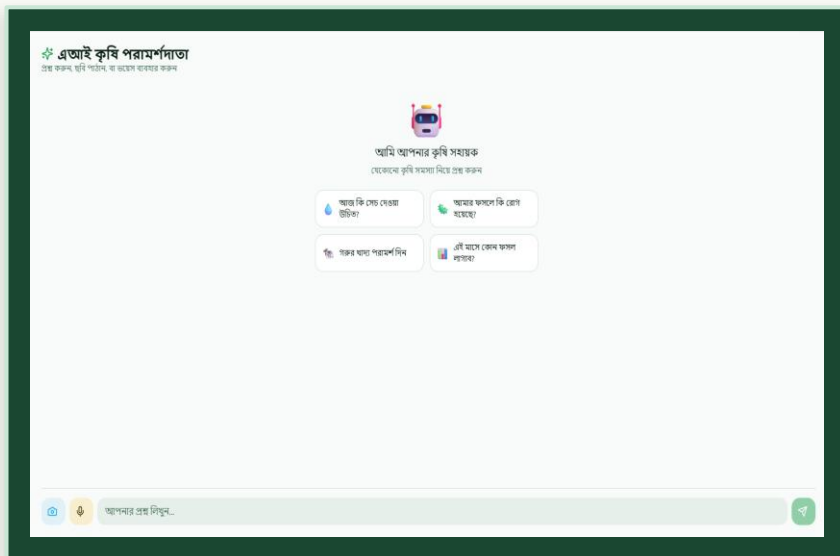
Fertilizer & irrigation plans tailored to field conditions



Harvest timing guidance to maximize yield quality



AI Dashboard



02

Low-Power Sensor Network

Affordable, solar-powered sensor kits deployed at the farm level — continuously monitoring soil conditions and local microclimate without requiring a power grid.



Soil Moisture

Real-time



Nutrient Levels

NPK Tracking



Microclimate

Field Forecast



Solar Charging

Off-grid Ready

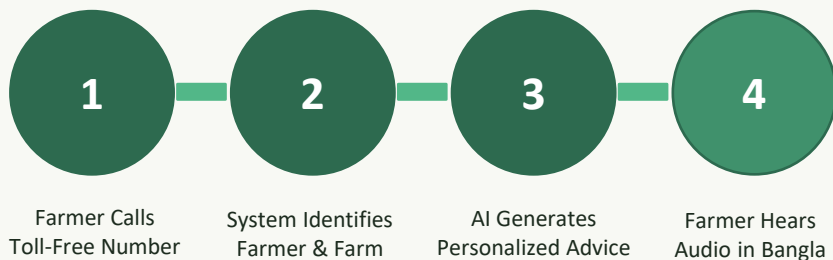


Sensor View



03 Voice-First Farmer Interface

A farmer dials a toll-free number and hears personalized advice in Bangla no smartphone, no literacy, no internet needed. Two-way voice queries are supported.



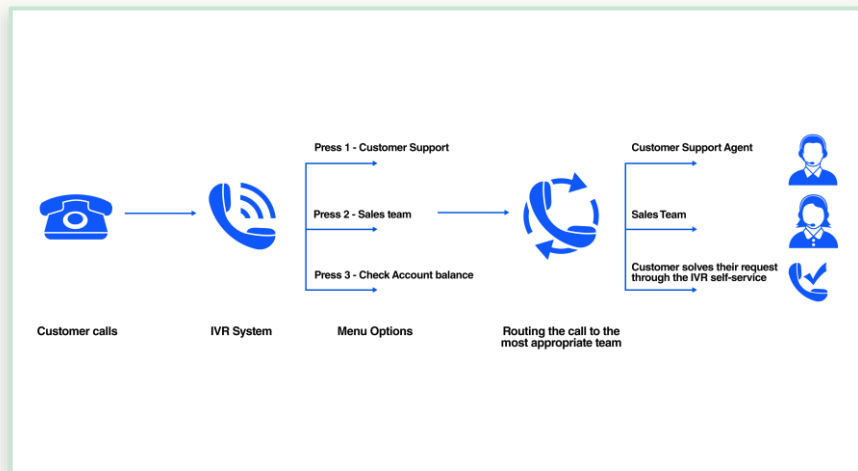
✓ No smartphone needed

✓ Works on basic phones

✓ Audio in Bangla dialect

✓ 80% accessibility rate

IVR Flow



Why KrishiOS is Unique



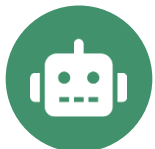
Low-Literacy First

IVR and voice UX specifically designed for non-readers and first-time tech users.



Solar-Powered

Energy autonomous devices for off-grid deployment across remote rural regions.



Local-Language AI

Models trained specifically on regional crops, dialects, and farming patterns in Bangladesh.



Offline Resilience

Gateway sync and local caching ensures reliable operation with intermittent networks.

vs Traditional: No internet required · Solar-powered · Field-level AI · Works on basic phones · Offline-first sync

Market Opportunity

11M+

Smallholder Farmers
in Bangladesh

Primary target users

400M+

Farmers in South Asia

Potential regional scale

\$18B

Smart Agri Market
(South Asia, 2026)

Rapidly growing sector



High-volume user base with institutional buyers (govt, NGOs)



Recurring revenue via SaaS analytics & device subscriptions



Scalable model for neighboring countries with similar farming systems

11M+

Bangladesh
Farmers

400M+

South Asia
Farmers

\$18B

Smart Agri
Market

Business Model

Farmers are users; institutions & commercial partners are the primary paying customers, enabling inclusive pricing.



Government Partnerships

Deploy at scale via agricultural extension programs and subsidies.



NGO Collaborations

Field programs co-funded for food security and climate resilience.



Premium Analytics

Subscription dashboards for advanced insights and bulk data access.



Device Sales

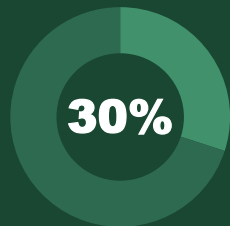
Low-cost sensor kits are sold through micro-distribution networks.



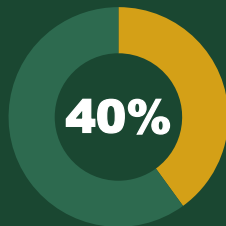
Agri-Company Integrations

Input suppliers and buyers pay for targeted analytics and lead gen.

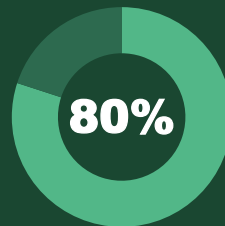
Impact & Future Vision



Estimated yield increase
per season



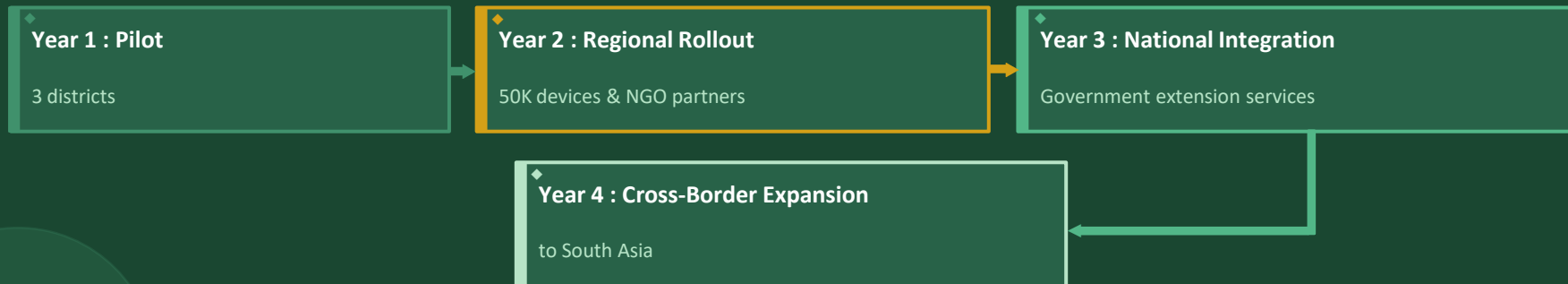
Reduction in preventable
crop losses



Accessibility for non-smartphone
users via IVR

Social impact: increased farmer incomes, improved food security, and climate-resilient practices powered by renewable energy.

Roadmap to Scale



Our Vision:

A nationwide, AI-powered farming network that amplifies smallholder resilience and creates data-driven agricultural policy insights.

Let's Build the Future of Agriculture Together

Team Krishi OS

Agri-tech engineers · Data scientists · Field operations leads

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 WebApp Link: <https://krishios.netlify.app/>

 Bangladesh